

Maintenance Manual — Aurora Petrochemical Refinery

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Plant	Aurora Petrochemical Refinery	Effective Date	15 Jan 2026
Document Owner	R. Boudreaux, Reliability Engineering Lead	Classification	Controlled — Site Use

1. Purpose & Scope

This manual defines preventive and predictive maintenance for rotating equipment, fired heaters, and pressure relief devices in Operating Units 200, 300, and 400 at Aurora Petrochemical Refinery. Applies to all maintenance employees and approved third-party turnaround contractors.

2. Critical Asset Register

The table below lists the critical assets covered by this manual. Each asset has a detailed task package in the CMMS (SAP-PM / Maximo) linked to its asset ID.

Asset ID	Description	Maintenance Strategy
P-201A/B	Crude charge pumps (centrifugal)	Monthly vibration, quarterly oil sample, annual mechanical seal inspection.
E-305	Hydrotreater feed-effluent exchanger	5-year cleaning per RBI plan; intermediate UT thickness scans.
H-401	Reformer fired heater	Daily tube IR thermography; biennial intelligent pigging of coil.
PSV-set-300	Pressure relief valves (Unit 300)	5-year pop-test cycle per API 510 inspection program.
C-202	Crude column overhead compressor	Online vibration monitoring; minimum monthly walkdown.

3. PM Schedule Summary

Frequency	Activity
Daily	Operator rounds, lube oil level check, gland leak inspection.
Weekly	Bearing temperature trending; thermography of high-criticality coil sections of H-401.
Monthly	Vibration spectra captured and uploaded to the CMMS; alarms triaged within 48 hours.
Annual	Mechanical seal swap on critical pumps; PSV documentation review per API 510.
Turnaround (5 yr)	Full inspection per RBI plan; pressure test; column tray inspection.

4. Work Management

All maintenance work is authorized through a CMMS work order. Work orders are categorized as PM, PdM, Corrective, Emergency, or Project. Maintenance KPIs (Schedule Compliance ≥85%, PM Compliance ≥95%, MTBF) are reported weekly to the Plant Manager.

5. Procedures

5.1 Work Permit Integration

All maintenance tasks require a work order in SAP-PM linked to a valid hot-work, cold-work, or confined-space permit. No tool starts work without the permit hard copy on site.

5.2 Reliability Data

Vibration, lube analysis, and thermography data flow into the AVEVA PI historian. Predictive alerts drive corrective work orders that bypass the standard backlog queue.

5.3 Bad Actor Program

Equipment with more than three unplanned failures in 12 months is reviewed by the Reliability Engineering team and assigned a Root Cause Failure Analysis with 30-day completion target.

6. References

- API 510 — Pressure Vessel Inspection Code
- API 570 — Piping Inspection Code
- API 581 — Risk-Based Inspection
- Internal: REL-APR-PROG-01 Reliability Program, RCFA-APR-002

7. Revision History

Rev	Date	Author	Summary
1	04 Jun 2023	R. Boudreaux	Initial issue.
2	21 Nov 2024	R. Boudreaux	Added predictive analytics integration.
3	09 Jul 2025	R. Boudreaux	Updated asset register.
4	15 Jan 2026	R. Boudreaux	Refreshed PM frequencies and KPIs.

Approvals

Role	Name	Signature	Date
Maintenance Engineering	R. Boudreaux, Reliability Engineering Lead	_____	15 Jan 2026
Reliability Engineering	Sr. Reliability Engineer	_____	15 Jan 2026
Operations / Site Lead	T. Whitfield, Operations Director	_____	15 Jan 2026

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